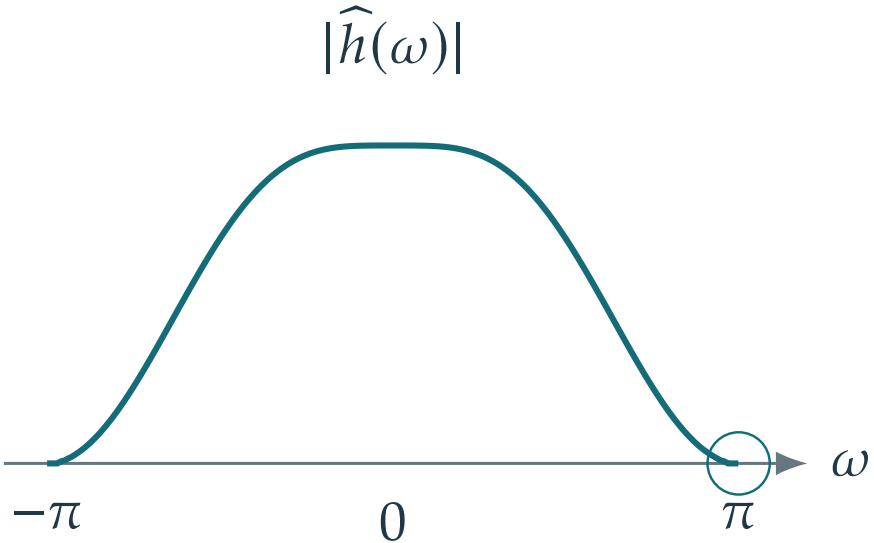
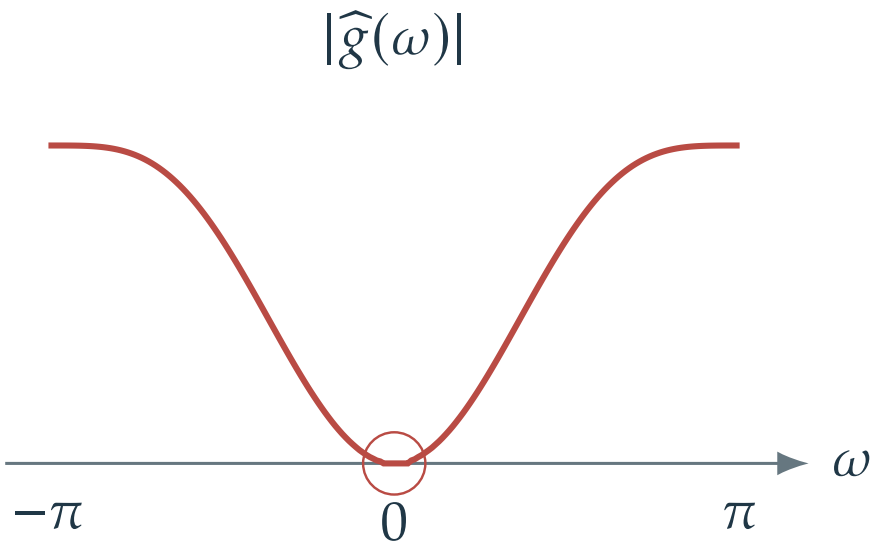
two vanishing moments illustrated ($p = 2$)




order-$p$ zero at $\pi$ order-$p$ zero at $0$

$$\widehat{h}^{(k)}(\pi) = \widehat{g}^{(k)}(0) = 0 \quad \Longleftrightarrow \quad \int_{\mathbb{R}} x^k \psi(x)\,\mathrm{d}x = 0, \qquad 0 \le k < p$$

Curves show magnitudes; derivatives refer to the filter symbols.